In [1]:
```
documents = {
    "Document 1": "The quick brown fox jumped over the lazy dog.",
    "Document 2": "The lazy dog slept in the sun."
}

for name, text in documents.items():
    print(name + ":", text)
```

```
Document 1: The quick brown fox jumped over the lazy dog.
Document 2: The lazy dog slept in the sun.
```

In [2]:
```
import re

tokenized_documents = {}

for name, text in documents.items():
    tokenized_documents[name] = re.findall(
        r"\b\w+\b", text.lower()
    )

print("Tokenized documents:")

for name, words in tokenized_documents.items():
    print(name, "->", words)
```

```
Tokenized documents:
Document 1 -> ['the', 'quick', 'brown', 'fox', 'jumped', 'over', 'the', 'lazy', 'do
g']
Document 2 -> ['the', 'lazy', 'dog', 'slept', 'in', 'the', 'sun']
```

In [3]:
```
inverted_index = {}

for document_name, words in tokenized_documents.items():
    for word in words:
        if word not in inverted_index:
            inverted_index[word] = []

        if document_name not in inverted_index[word]:
            inverted_index[word].append(document_name)
```

In [4]:
```
print("Inverted index:")

for word in sorted(inverted_index):
    print(word, "->", inverted_index[word])
```

```
Inverted index:
brown -> ['Document 1']
dog -> ['Document 1', 'Document 2']
fox -> ['Document 1']
in -> ['Document 2']
jumped -> ['Document 1']
lazy -> ['Document 1', 'Document 2']
over -> ['Document 1']
quick -> ['Document 1']
slept -> ['Document 2']
sun -> ['Document 2']
the -> ['Document 1', 'Document 2']
```

In [5]:
```
search_word = input("Enter a word to search: ").strip().lower()

matching_documents = inverted_index.get(search_word, [])

if matching_documents:
    print("Documents containing", repr(search_word) + ":")

    for document_name in matching_documents:
        print(document_name, "->", documents[document_name])
else:
    print("No documents found for", repr(search_word))
```

```
Documents containing 'jumped':
Document 1 -> The quick brown fox jumped over the lazy dog.
```